Project: Worked examples
Title: Accumulator sizing (gas law)
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1. Scope and method

Calculates the pre-charge gas volume V0 of a hydraulic accumulator that delivers or absorbs a liquid volume ΔV between two working pressures, with the next listed size and the PSSR bar·litre product. It is not a pressure-vessel code check.

WHAT IT CALCULATES	RELATION
Absolute pressures	$p_a = p_{gauge} + p_{atm}$, or as entered in Absolute mode
Gas volume, one exponent both ways	$V0 = \Delta V / [(p0/p1)^{(1/n_{gas})} - (p0/p2)^{(1/n_{gas})}]$, n_{gas} = 1 isothermal to 1.4 adiabatic
Gas volume, slow charge then discharge at n_{gas}	$V0 = \Delta V \cdot (p2/p0) / [(p2/p1)^{(1/n_{gas})} - 1]$, from $V2 = V0 \cdot p0/p2$ and $V1 = V2 \cdot (p2/p1)^{(1/n_{gas})}$
Temperature and gas corrections	$V0 \times f_{rg} \times$ the cold-gas factor T_{max}/T_{min}, to the power $1/n_{gas}$ when one exponent holds both ways, in absolute temperatures
Pre-charge rules	$p0 \le r_0 \cdot p1$ at T_{max}, and $p2/p0(T_{min}) \le r_p$ with $p0(T_{min}) = p0 \cdot T_{min}/T_{max}$ in absolute temperatures
Pre-charge at the charging temperature	$p0(T_c) = p0(T_{max}) \cdot T_c/T_{max}$ in absolute temperatures
Next size	the smallest listed size not below V0
PSSR bar-litre product	PS·V of the vessel against 250 bar·L, an indication that needs PS and a volume

Basis: 250 bar·litre exception, indication only[1](§Schedule 1); Pressure equipment, not checked here[2]; Gas-charged accumulator sizing, $p \cdot V^n = const$[3](Ch. on accumulators, 7th ed.).

2. Inputs

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
mode	Duty	Energy storage		assumption: Slow clamp-and-hold cycle: stored-energy duty.	any
process	Gas process (By duty: an emergency stroke is charged slowly to p_2 and discharged at n_{gas}, the other duties use one exponent both ways)	By duty		default	any
p_{basis}	Pressures entered as	Absolute		assumption: Pressures entered as absolute bar, as the source numbers are quoted.	any

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
dV	Liquid volume delivered, absorbed or needed for the emergency stroke	14	L	spec: Hydraulic specification HS-12: clamp stroke plus leakage make-up, 14 L per cycle	0.01 … 500 L
p_0	Gas pre-charge pressure p0, at the maximum operating temperature	90	bar a	spec: Hydraulic specification HS-12: pre-charge 90 bar at the maximum oil temperature	1 … 400 bar a
p_1	Lowest working pressure p1	100	bar a	spec: Hydraulic specification HS-12: lowest clamp pressure 100 bar	1 … 400 bar a
p_2	Highest working pressure p2	210	bar a	spec: Hydraulic specification HS-12: pump cut-out 210 bar	1 … 400 bar a
p_{atm}	Atmospheric pressure (gauge to absolute)	1.01325	bar a	assumption: Standard atmosphere, derived constant.	0.7 … 1.05 bar a
T_{max}	Maximum operating temperature (p0 applies here)	50	°C	spec: Hydraulic specification HS-12: oil temperature limit 50 °C	-20 … 120 °C
T_{min}	Minimum operating temperature (the gas is coldest and delivers least here)	30	°C	assumption: Oil temperature range assumed 30 to 50 °C in service (invented for this example); the gas is coldest at 30 °C.	-30 … 100 °C
T_c	Charging temperature	20	°C	assumption: Charged in the workshop at 20 °C.	-20 … 60 °C
n_{gas}	Gas exponent of the discharge, 1 isothermal (slow) to 1.4 adiabatic (fast), also the charge exponent when one exponent holds both ways	1		assumption: Isothermal: the clamp holds for minutes between cycles.	1 … 1.4
f_{rg}	Real-gas factor on V0, the real over the ideal gas volume for this pressure window (1 = ideal gas, at least 1)	1		assumption: Ideal gas; the highest pressure is just above 200 bar, so the sheet warns.	1 … 2.1

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
r_0	Pre-charge rule: p0 (at T_{max}) at most this fraction of p1, on absolute pressures	0.9		reference: record: accumulator pre-charge rules (bladder type), to be taken from the manufacturer's guide	0.5 ... 0.95
r_p	Pre-charge rule: p2/p0 (p0 at T_{min}) at most this ratio, on absolute pressures	4		reference: record: accumulator pre-charge rules (bladder type), to be taken from the manufacturer's guide	2 ... 10
sizes	Available accumulator sizes from the manufacturer's range (optional; blank = no size is selected)	4 row(s)		reference: record: illustrative size list invented for this example, to be replaced by the manufacturer's range	any
PS	Vessel maximum allowable pressure, bar gauge, from the nameplate or relief setting (optional; blank = no PSSR indication)	230	bar g	spec: Hydraulic specification HS-12: accumulator relief setting 230 bar gauge (vessel PS; illustrative, to be replaced by the nameplate)	10 ... 700 bar g

TABLE ROW	VALUES
sizes[1]	V = 10 L
sizes[2]	V = 20 L
sizes[3]	V = 32 L
sizes[4]	V = 50 L

3. Assumptions

A1. All gas-law pressures are absolute. ($p \cdot V^n$ = const on a fixed mass of gas.)
A2. A gauge entry is converted with the atmospheric pressure entered on the sheet. ($p \cdot V^n$ = const on a fixed mass of gas.)
A3. The exponent n_{gas} is yours, from 1 for a slow isothermal discharge to 1.4 for a fast adiabatic one. (Gas-law exponents, closed form.)
A4. With one exponent both ways, a larger n_{gas} gives a larger volume for a given ΔV. (Gas-law exponents, closed form.)
A5. A held accumulator is charged slowly and isothermally to p_2, then discharged at n_{gas}. (Gas-law sizing of a held accumulator: slow charge, fast discharge.)
A6. An emergency stroke is a held accumulator by default. (Gas-law sizing of a held accumulator: slow charge, fast discharge.)
A7. Sizing a held accumulator with one exponent understates the volume. (Gas-law sizing of a held accumulator: slow charge, fast discharge.)

A8. The one-exponent form suits a gas charged as fast as it is discharged. (Gas-law sizing of a held accumulator: slow charge, fast discharge.)
A9. The shock and pulsation duty always uses one exponent. (Gas-law sizing of a held accumulator: slow charge, fast discharge.)
A10. The gas is ideal unless a real-gas factor is entered. (Real-gas correction as a user factor.)
A11. The real-gas factor is at least 1, because real nitrogen needs more volume than ideal gas, never less. (Real-gas correction as a user factor.)
A12. Above 200 bar the sheet warns that nitrogen departs from the ideal-gas law. (Real-gas correction as a user factor.)
A13. The pre-charge p0 is defined at the maximum operating temperature T_{max}, and V0 is sized there. (Manufacturer sizing practice and the gas law at constant mass.)
A14. At T_{min} the same gas delivers only the fraction T_{min}/T_{max} of ΔV in absolute temperatures. (Manufacturer sizing practice and the gas law at constant mass.)
A15. A held accumulator multiplies V0 by T_{max}/T_{min}. (Manufacturer sizing practice and the gas law at constant mass.)
A16. One exponent n_{gas} both ways multiplies V0 by T_{max}/T_{min} to the power $1/n_{gas}$. (Manufacturer sizing practice and the gas law at constant mass.)
A17. The p2/p0 rule is checked with the pre-charge at T_{min}. (Manufacturer sizing practice and the gas law at constant mass.)
A18. With T_{min} equal to T_{max} the temperature correction is 1. (Manufacturer sizing practice and the gas law at constant mass.)
A19. The pre-charge rules are records for the accumulator type and make. (Rules are records, not built-in limits.)
A20. The rules are checked on absolute pressures, whichever basis the pressures are entered on. (Rules are records, not built-in limits.)
A21. The manufacturer gives the rule values and says whether its ratio is on gauge or absolute pressures. (Rules are records, not built-in limits.)
A22. The bar·litre product is PS times the vessel volume V. (Regulatory gating: no pressure-vessel code check.)
A23. PS is the vessel's maximum allowable pressure in bar gauge, never below the highest working pressure p_2. (Regulatory gating: no pressure-vessel code check.)
A24. The vessel volume is the selected size unless you enter it. (Regulatory gating: no pressure-vessel code check.)
A25. Without PS and a volume the PSSR indication is not available. (Regulatory gating: no pressure-vessel code check.)
A26. The indication shows whether the 250 bar·L exception can apply, and it is not a compliance statement. (Regulatory gating: no pressure-vessel code check.)
A27. The PED, UK PER, PSSR examination and the vessel's design code stay with the manufacturer and the competent person. (Regulatory gating: no pressure-vessel code check.)
A28. mode (Duty) is taken as Energy storage. (Slow clamp-and-hold cycle: stored-energy duty.)
A29. p_{basis} (Pressures entered as) is taken as Absolute. (Pressures entered as absolute bar, as the source numbers are quoted.)
A30. p_{atm} (Atmospheric pressure (gauge to absolute)) is taken as 1.01325 bar a. (Standard atmosphere, derived constant.)
A31. T_{min} (Minimum operating temperature (the gas is coldest and delivers least here)) is taken as 30 °C. (Oil temperature range assumed 30 to 50 °C in service (invented for this example); the gas is coldest at 30 °C.)
A32. T_c (Charging temperature) is taken as 20 °C. (Charged in the workshop at 20 °C.)
A33. n_{gas} (Gas exponent of the discharge, 1 isothermal (slow) to 1.4 adiabatic (fast), also the charge exponent when one exponent holds both ways) is taken as 1. (Isothermal: the clamp holds for minutes between cycles.)
A34. f_{rg} (Real-gas factor on V0, the real over the ideal gas volume for this pressure window (1 = ideal gas,

at least 1)) is taken as 1. (Ideal gas; the highest pressure is just above 200 bar, so the sheet warns.) A35. Energy storage duty: ΔV is the liquid volume delivered between p_2 and p_1 over a cycle, with n_{gas} from 1 for a slow discharge to 1.4 for a fast one. (Gas-law sizing for a stored-energy duty.)

4. Working

[15] $pa_0 = p_0$
= 90 bar a
= 90 bar a
Pre-charge pressure, absolute

[16] $pa_1 = p_1$
= 100 bar a
= 100 bar a
Lowest working pressure, absolute

[17] $pa_2 = p_2$
= 210 bar a
= 210 bar a
Highest working pressure, absolute

[18] $p2_g = \max(pa_2 - p_{atm}, 0\ bar)$
= max(210 bar a − 1.0132 bar a, 0 bar)
= 208.99 bar g
Highest working pressure, gauge

[21] $V0_{ideal} = dV / [(pa_0/pa_1)^{(1/n_{gas})} - (pa_0/pa_2)^{(1/n_{gas})}]$ [4](const)
= 14 L / [(0.9) − (0.42857)]
= 29.697 L
Gas volume for the displacement, ideal gas

[22] $f_{Tabs} = (T_{max} / 1\ K) / (T_{min} / 1\ K)$
= 50 degC / 1 K / (30 degC / 1 K)
= 1.066
Absolute temperature ratio T_{max}/T_{min}

[23] $f_T = f_{Tabs}^{(1 / n_{gas})}$
= 1.066^(1 / 1)
= 1.066
Temperature scaling of the gas volume: the temperature ratio to the power 1/n (one exponent both ways)

[24] $V0 = V0_{ideal} * f_{rg} * f_T$
= 29.697 L × 1 × 1.066
= 31.656 L
Pre-charge gas volume V0 (accumulator size required)

[25] $T_{ratio} = (T_c / 1\ K) / (T_{max} / 1\ K)$
= 20 degC / 1 K / (50 degC / 1 K)
= 0.90716
Charging over maximum absolute temperature

[26] $p0 = pa_0 * T_{ratio}$
= 90 bar a × 0.90716
= 81.645 bar a
Absolute pre-charge pressure at the charging temperature

[27] $p0_{min} = pa_0 / f_{Tabs}$
= 90 bar a / 1.066
= 84.43 bar a
Absolute pre-charge pressure at the minimum operating temperature

[28] $p0_{set} = p0$
= 81.645 bar a
= 81.645 bar a
Pre-charge to set at the charging temperature (absolute)

[29] $p0_{max} = r_0 * pa_1$
= 0.9 × 100 bar a
= 90 bar a
Largest pre-charge the p0/p1 rule allows
[30] $u_{r0} = pa_0 / p0_{max}$
= 90 bar a / 90 bar a
= 1
Pre-charge against the p0/p1 rule
[31] $p_{ratio} = pa_2 / p0_{min}$
= 210 bar a / 84.43 bar a
= 2.4873
Pressure ratio p2/p0, with p0 at the minimum temperature
[32] $u_{rp} = p_{ratio} / r_p$
= 2.4873 / 4
= 0.62182
Pressure ratio p2/p0 (p0 at the minimum temperature) against the rule
[33] V_{sel} = smallest listed size not below V0
= V0 = 31.656 L, 4 sizes listed
= 32 L
Selected accumulator size
[34] $u_V = V0 / V_{sel}$
= 31.656 L / 32 L
= 0.98926
Required volume over the selected size
[35] $bar_{litres} = PS * V_{sel}$
= 230 bar g × 32 L
= 7360 bar·L
Pressure-volume product PS·V (PS as entered, V the selected size)
[36] BL_{pssr} = 250 bar*L [5](1)
= 250 bar·L
PSSR Schedule 1 exception threshold (indication)

5. Results

SYMBOL	VALUE	DESCRIPTION
V0	31.66 L	Accumulator gas volume required
p0	81.64 bar a	Pre-charge, absolute, at the charging temperature
bar_{litres}	7360 bar·L	Pressure-volume product (bar·litres)
V_{sel}	32 L	Selected size
$p0_{set}$	81.64 bar a	Pre-charge to set at the charging temperature
$p0_{min}$	84.43 bar a	Pre-charge, absolute, at the minimum operating temperature
f_T	1.066	Gas volume scaling T_{max}/T_{min}
pssr	"7360 bar·L is at or above 250 bar·L: the exception does not apply; PSSR and PED / UK PER duties follow."	PSSR Schedule 1 250 bar·L exception (indication only)

SYMBOL	VALUE	DESCRIPTION
process	"charged as fast as it is discharged (one exponent n_{gas} = 1 from the pre-charge to both working pressures)"	Gas process the volume is sized for
u_V	0.99	Selected size utilisation
u_{r0}	1.00	Pre-charge rule utilisation (p0/p1) (governing)
u_{rp}	0.62	Pressure ratio utilisation (p2/p0 at the minimum temperature)

6. Checks (governing first)

RESULT	CHECK	DEMAND	CAPACITY	UTILISATION	REF
PASS	Pre-charge against the p0/p1 rule	90 bar a	90 bar a	1.00	[6](record)
PASS	Pressure ratio against the p2/p0 rule	2.4873	4	0.62	[6](record)

7. Applicability

WARNING records: tabulated values not taken from a reference record are r_0, r_p (Pre-charge rules); sizes (Accumulator size list). Enter each as one so its source, edition and licence travel with it. [record]
WARNING p_2: the highest working pressure is 210 bar absolute, above 200 bar, where nitrogen departs from the ideal gas. The volume shown is too small, so enter f_{rg} (up to 3) from a gas-property source. [realgas]

Defaults used (inputs left empty):
process: not entered, so the default of 'By duty' is used.

8. Verification

```
Template fp.accumulator@1.7.0 Content hash 7ea8865ce36556d1 Engine calc-engine/1.0.0
Known-answer cases passed 10/10 Test run 2026-10-23 12:00 UTC TempestOS 0.26.0
```

9. References

1. Pressure Systems Safety Regulations 2000 (SI 2000/128).
2. Pressure Equipment Directive 2014/68/EU and UK PER 2016.
3. Esposito, Fluid Power with Applications.
4. p·Vⁿ =.
5. PSSR Schedule.
6. accumulator.rules.